

FACULTÉ DES SCIENCES ET TECHNOLOGIES (FST)

TD No 4-Administration Système

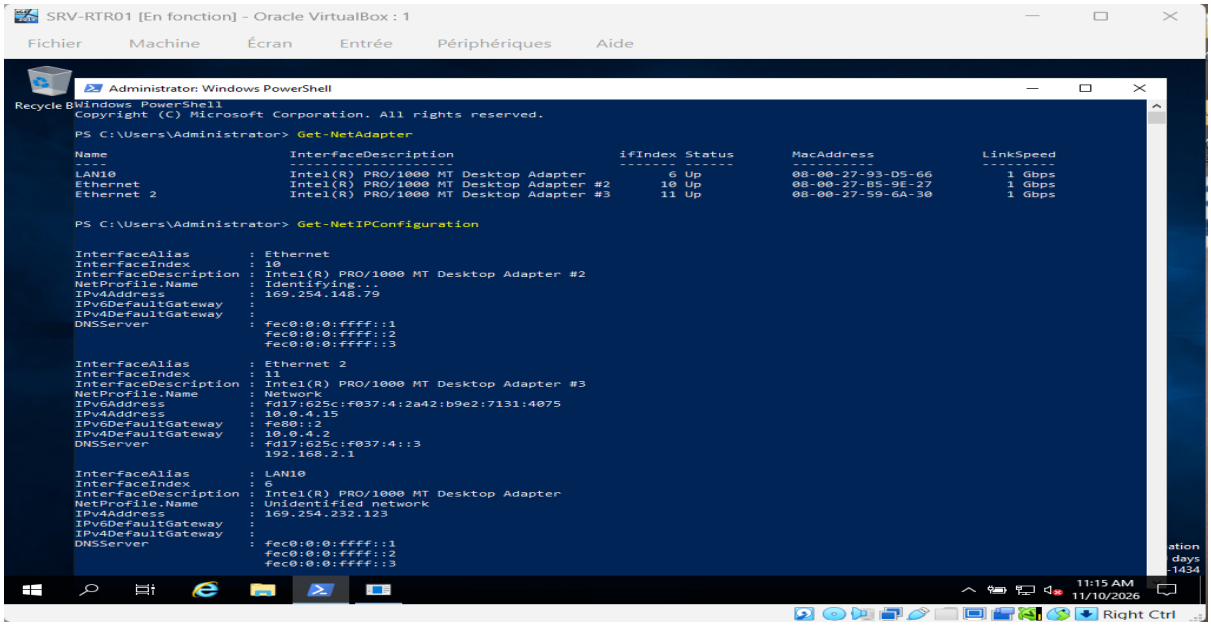
Devoir rédigé par : Ruth-Jose JEROME

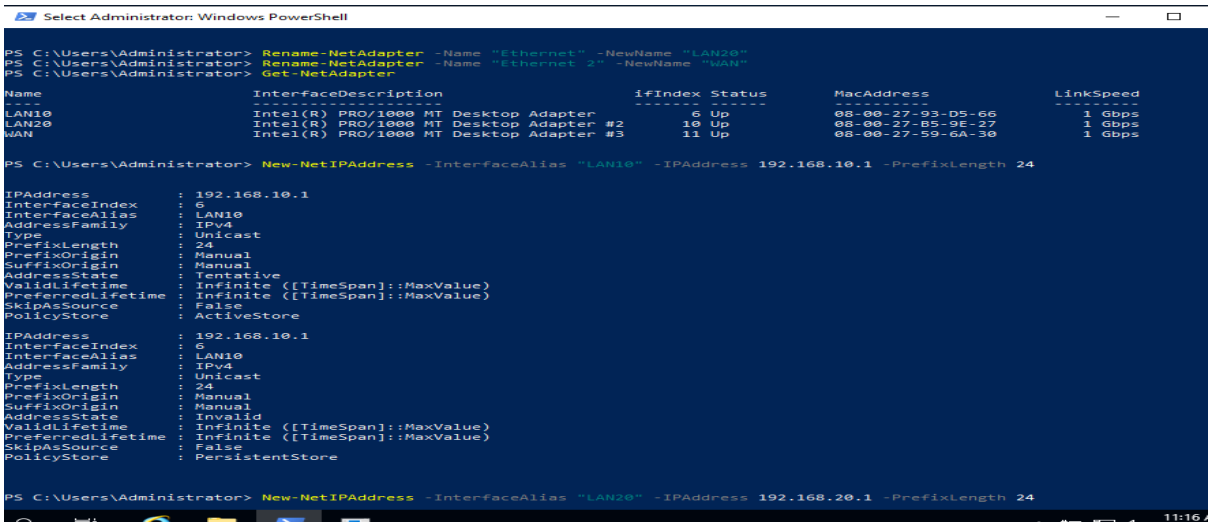
Étudiant en L4

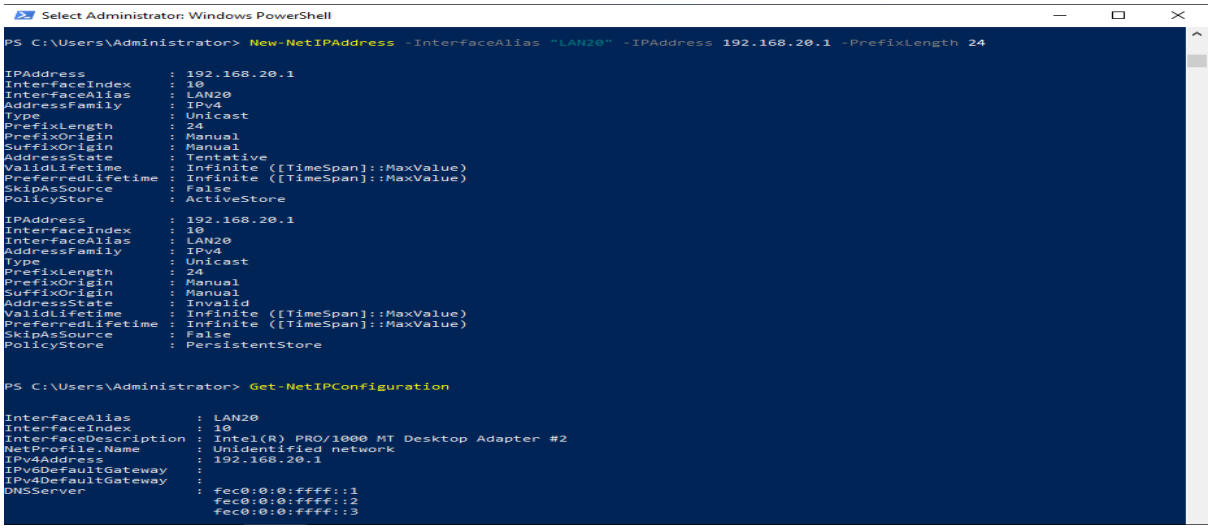
Soumis au professeur : Ismael SAINT AMOUR

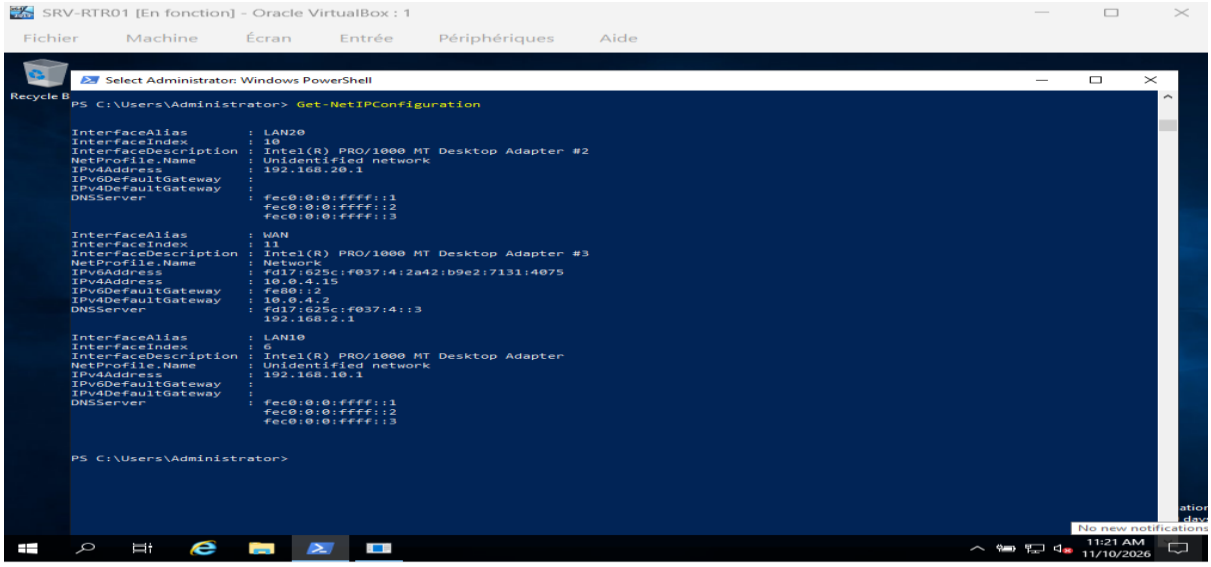
DATE: 09 octobre 2026

Activité 1 Préparer le routeur

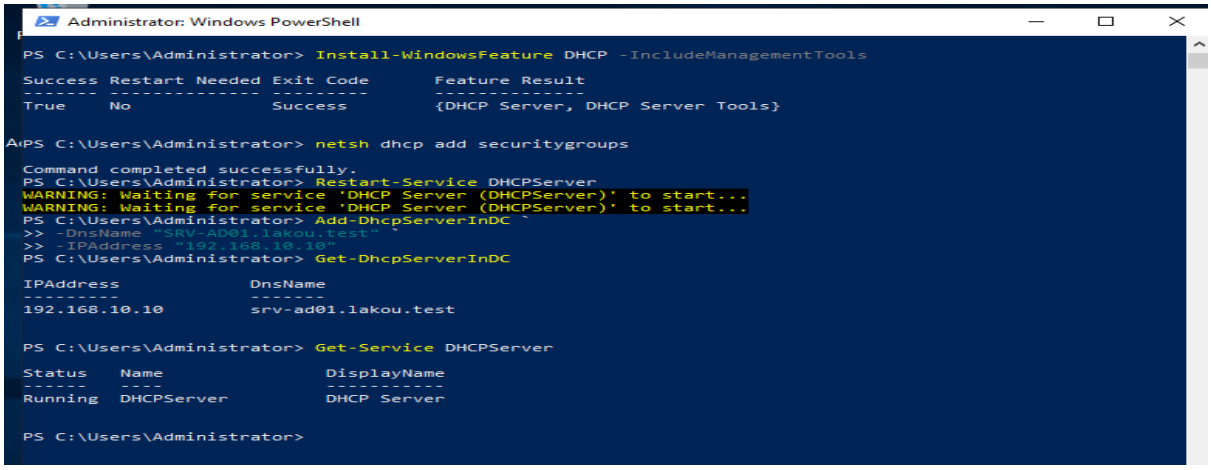




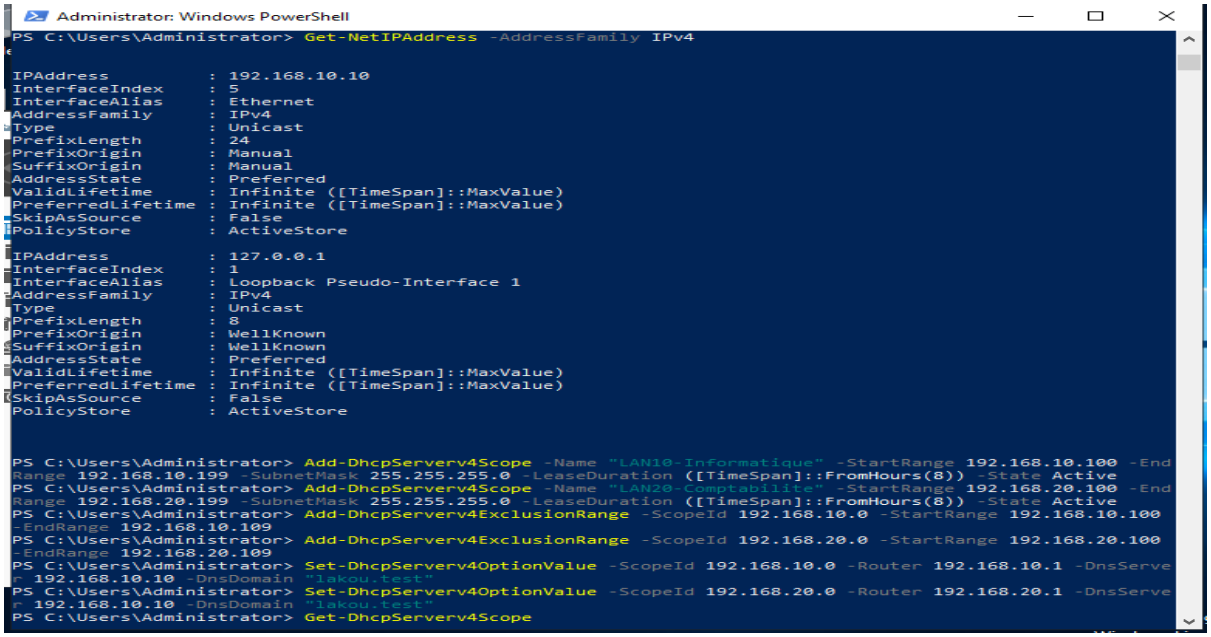


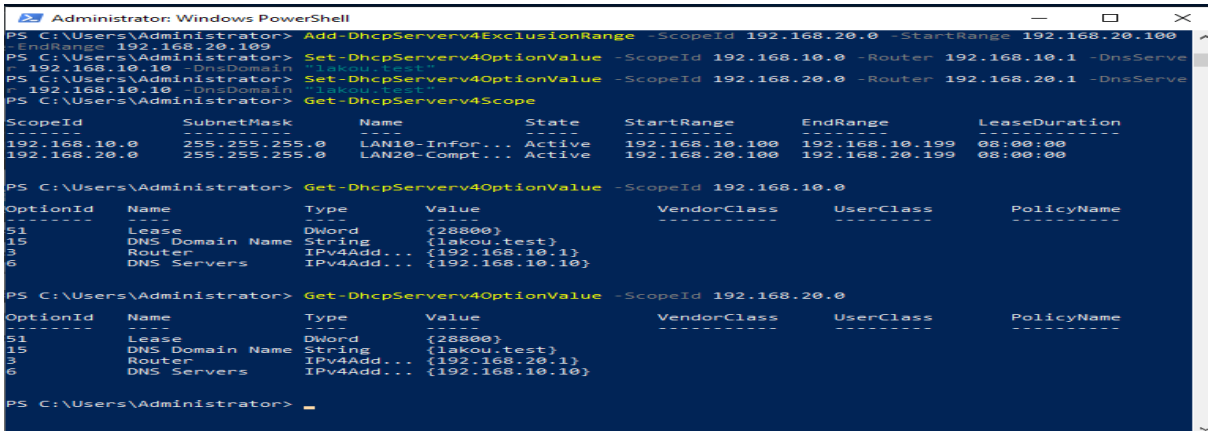


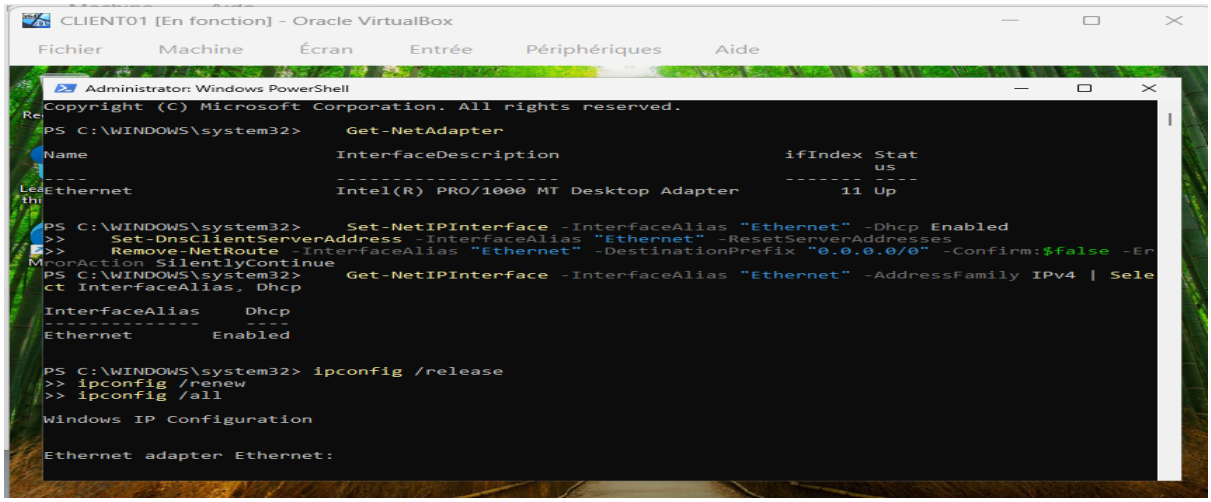
Activité 2 Installer et autoriser DHCP

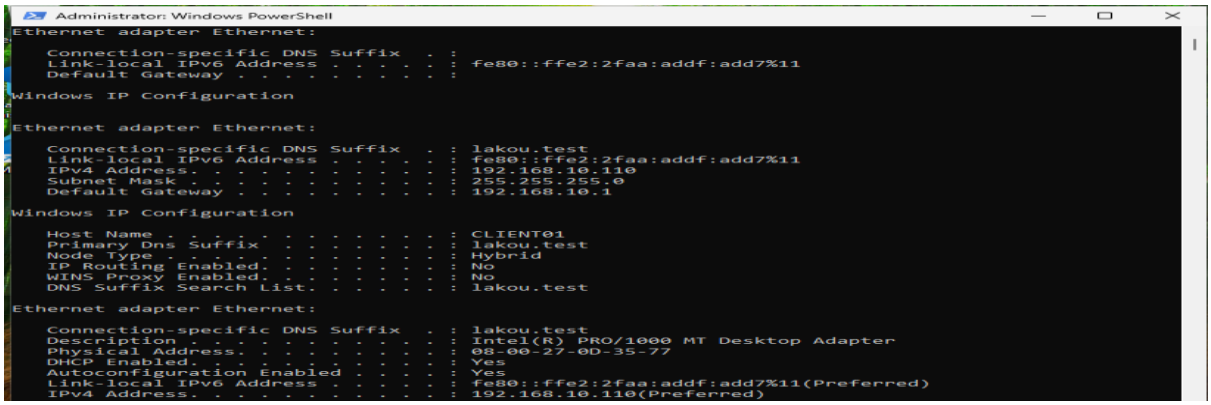


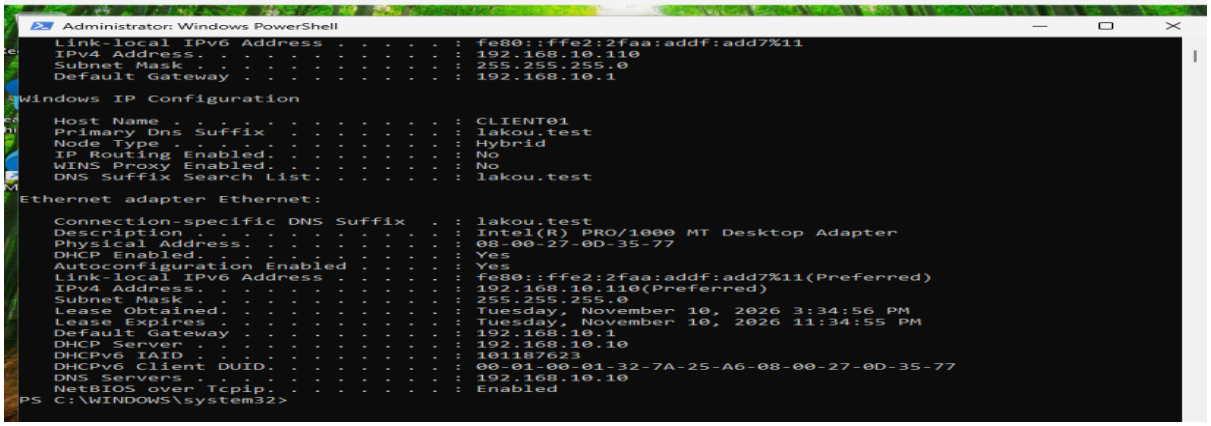
Activité 3 Créer les étendues DHCP



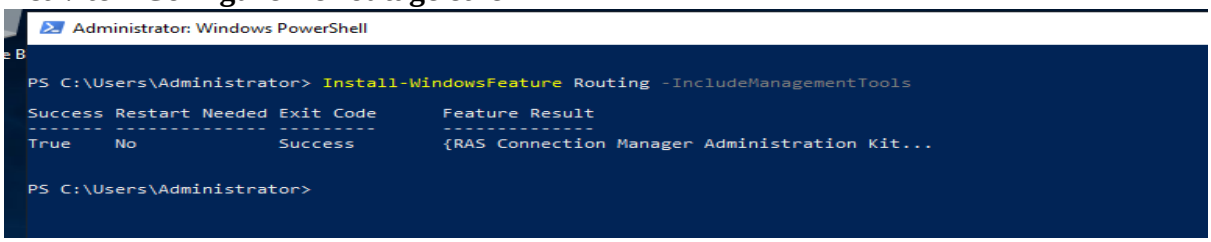

Administrator: Windows PowerShell
PS C:\Users\Administrator> Add-DhcpServerv4ExclusionRange -ScopeId 192.168.20.0 -StartRange 192.168.20.100 -EndRange 192.168.20.109
PS C:\Users\Administrator> Set-DhcpServerv4OptionValue -ScopeId 192.168.10.0 -Router 192.168.10.1 -DnsServer 192.168.10.10 -DnsDomain "lakou.test"
PS C:\Users\Administrator> Set-DhcpServerv4OptionValue -ScopeId 192.168.20.0 -Router 192.168.20.1 -DnsServer 192.168.10.10 -DnsDomain "lakou.test"
PS C:\Users\Administrator> Get-DhcpServerv4Scope
ScopeId SubnetMask Name State StartRange EndRange LeaseDuration
192.168.10.0 255.255.255.0 LAN10-Infor... Active 192.168.10.100 192.168.10.199 08:00:00
192.168.20.0 255.255.255.0 LAN20-Compt... Active 192.168.20.100 192.168.20.199 08:00:00
PS C:\Users\Administrator> Get-DhcpServerv4OptionValue -ScopeId 192.168.10.0
OptionId Name Type Value VendorClass UserClass PolicyName
51 Lease DWord {28800}
15 DNS Domain Name String {lakou.test}
3 Router IPv4Add... {192.168.10.1}
6 DNS Servers IPv4Add... {192.168.10.10}
PS C:\Users\Administrator> Get-DhcpServerv4OptionValue -ScopeId 192.168.20.0
OptionId Name Type Value VendorClass UserClass PolicyName
51 Lease DWord {28800}
15 DNS Domain Name String {lakou.test}
3 Router IPv4Add... {192.168.20.1}
6 DNS Servers IPv4Add... {192.168.10.10}
PS C:\Users\Administrator>

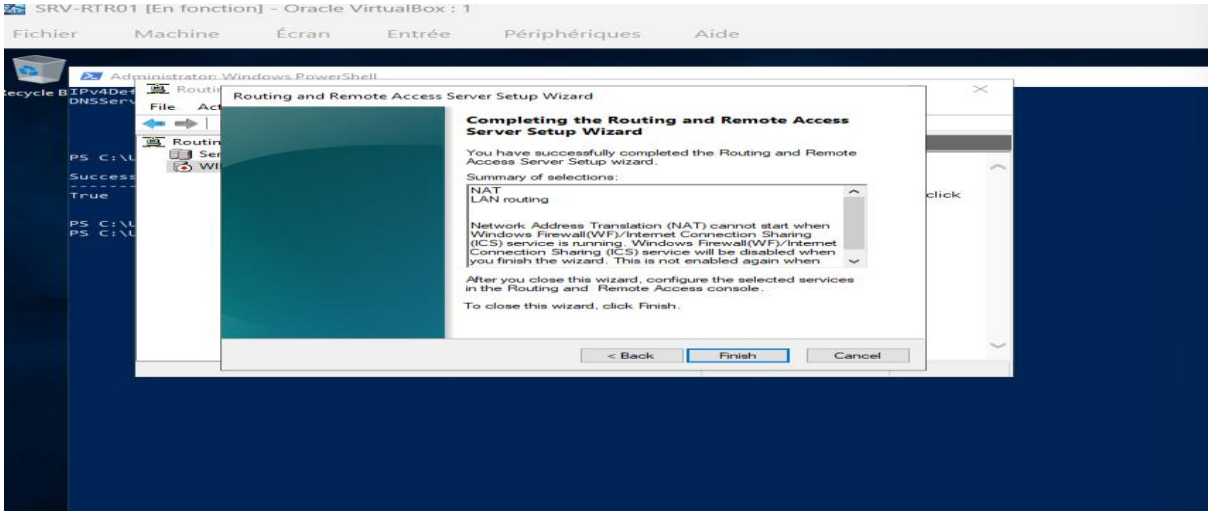

CLIENT01 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\WINDOWS\system32> Get-NetAdapter
Name InterfaceDescription ifIndex Status
Ethernet Intel(R) PRO/1000 MT Desktop Adapter 11 Up
PS C:\WINDOWS\system32> Set-NetIPInterface -InterfaceAlias "Ethernet" -Dhcp Enabled
>> Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ResetServerAddresses
>> Remove-NetRoute -InterfaceAlias "Ethernet" -DestinationPrefix "0.0.0.0/0" -Confirm:$false -ErrorAction SilentlyContinue
PS C:\WINDOWS\system32> Get-NetIPInterface -InterfaceAlias "Ethernet" -AddressFamily IPv4 | Select InterfaceAlias, Dhcp
InterfaceAlias Dhcp
Ethernet Enabled
PS C:\WINDOWS\system32> ipconfig /release
>> ipconfig /renew
>> ipconfig /all
Windows IP Configuration
Ethernet adapter Ethernet:

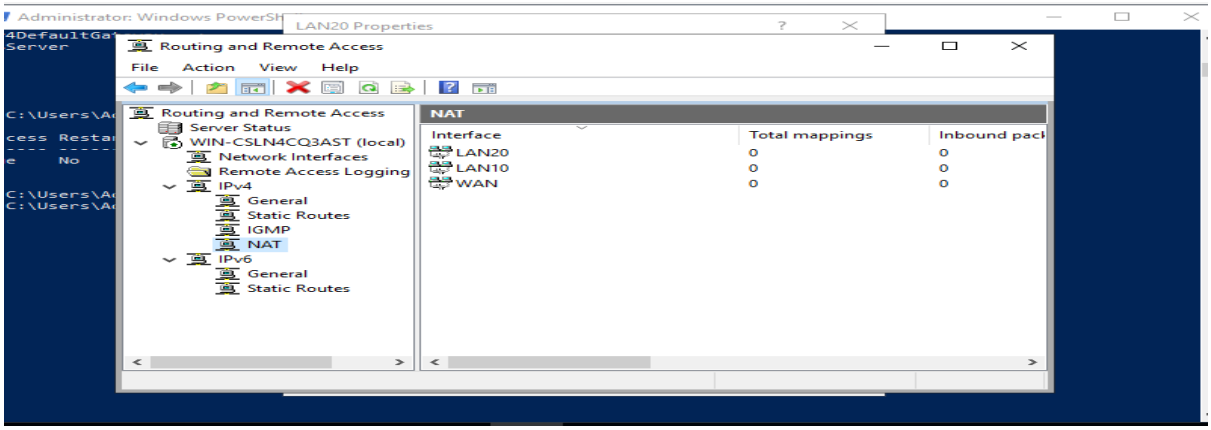

Administrator: Windows PowerShell
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::ffe2:2faa:addf:add7%11
Default Gateway . . . . . :
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : lakou.test
Link-local IPv6 Address . . . . . : fe80::ffe2:2faa:addf:add7%11
IPv4 Address. . . . . : 192.168.10.110
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.10.1
Windows IP Configuration
Host Name . . . . . : CLIENT01
Primary Dns Suffix . . . . . : lakou.test
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : lakou.test
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : lakou.test
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-0D-35-77
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::ffe2:2faa:addf:add7%11(Preferred)
IPv4 Address. . . . . : 192.168.10.110(Preferred)

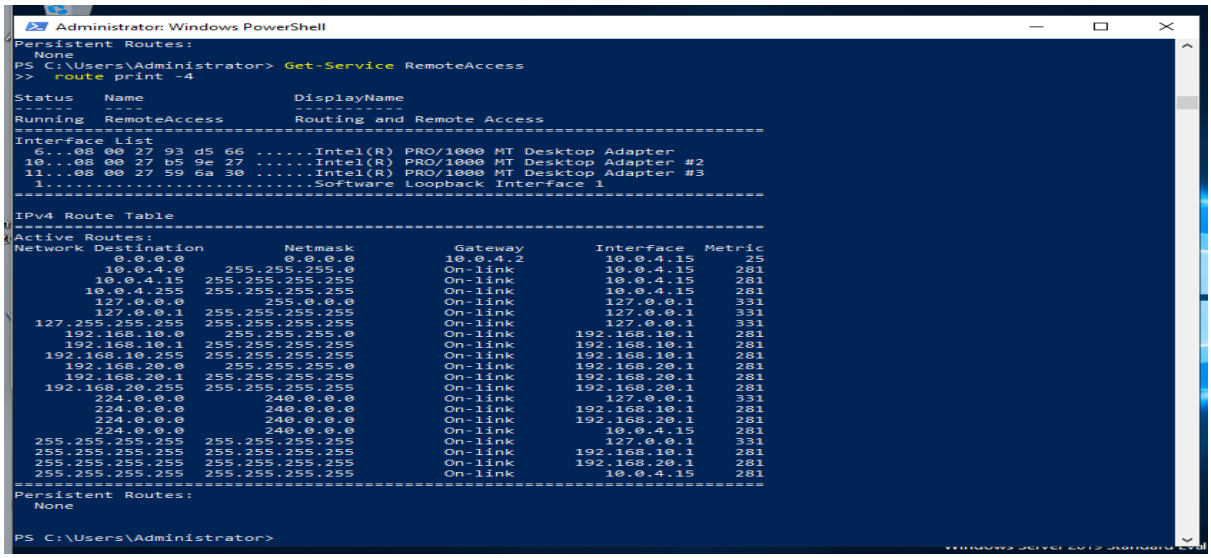

Administrator: Windows PowerShell
Link-local IPv6 Address . . . . . : fe80::ffe2:2faa:addf:add7%11
IPv4 Address. . . . . : 192.168.10.110
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.10.1
Windows IP Configuration
Host Name . . . . . : CLIENT01
Primary Dns Suffix . . . . . : lakou.test
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : lakou.test
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : lakou.test
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-0D-35-77
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::ffe2:2faa:addf:add7%11(Preferred)
IPv4 Address. . . . . : 192.168.10.110(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, November 10, 2026 3:34:56 PM
Lease Expires . . . . . : Tuesday, November 10, 2026 11:34:55 PM
Default Gateway . . . . . : 192.168.10.1
DHCP Server . . . . . : 192.168.10.10
DHCPv6 IAID . . . . . : 101187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-32-7A-25-A6-08-00-27-0D-35-77
DNS Servers . . . . . : 192.168.10.10
NetBIOS over Tcpip. . . . . : Enabled
PS C:\WINDOWS\system32>

Activité 4 Configurer le routage et le NAT







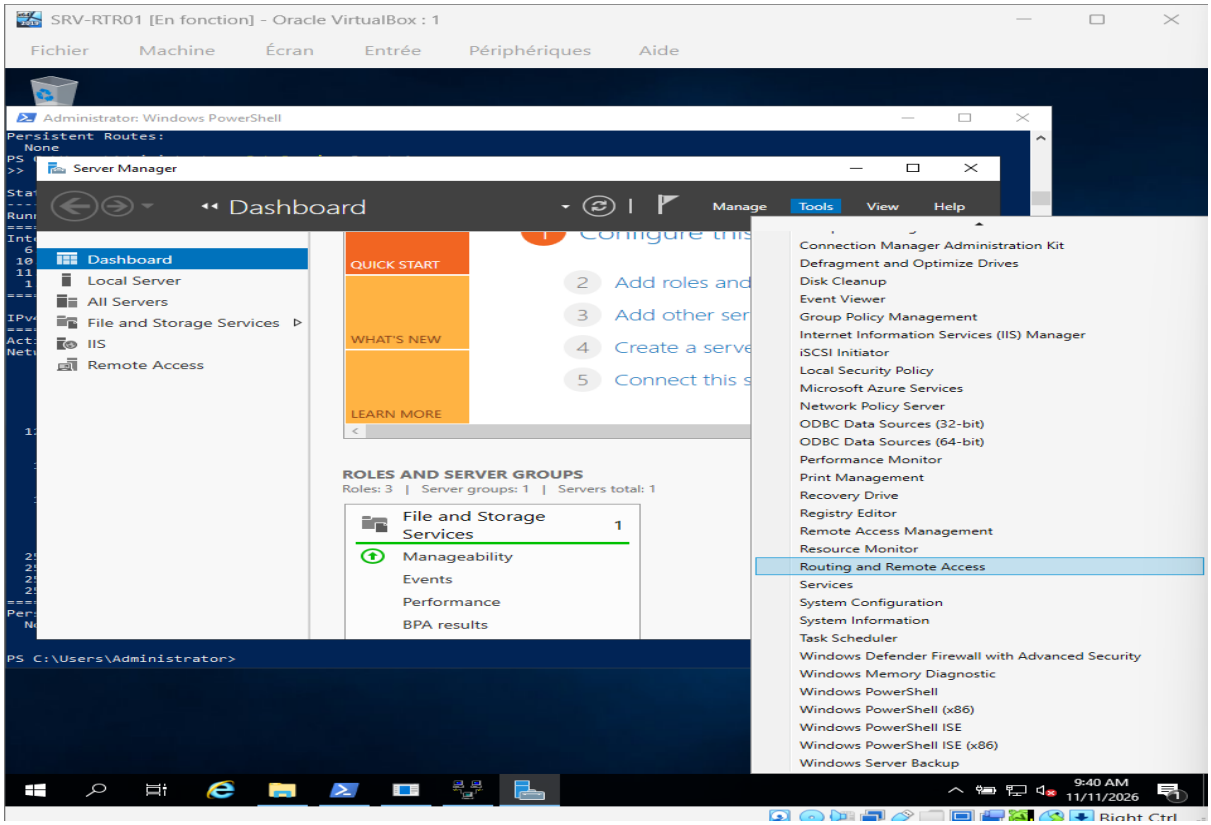


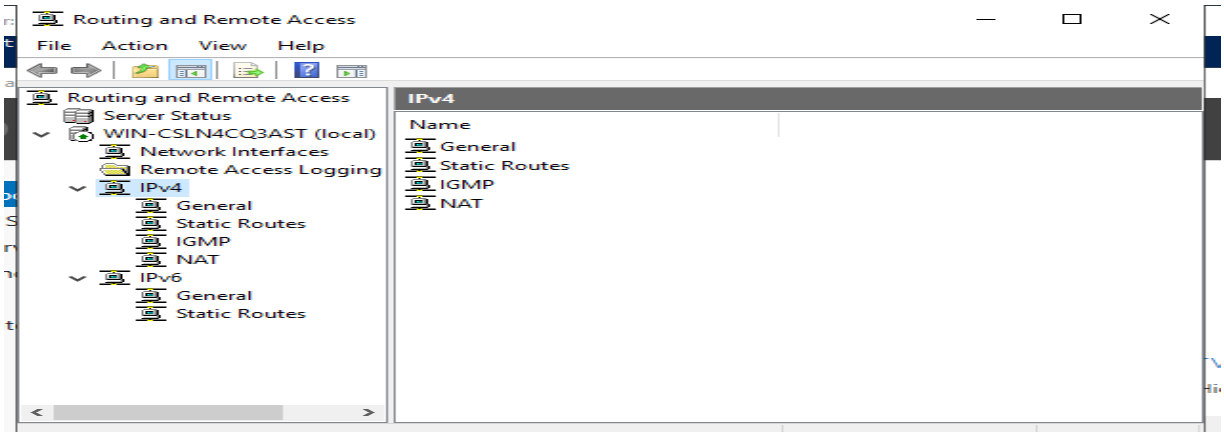
```
PS C:\Users\Administrator> Get-NetIPConfiguration

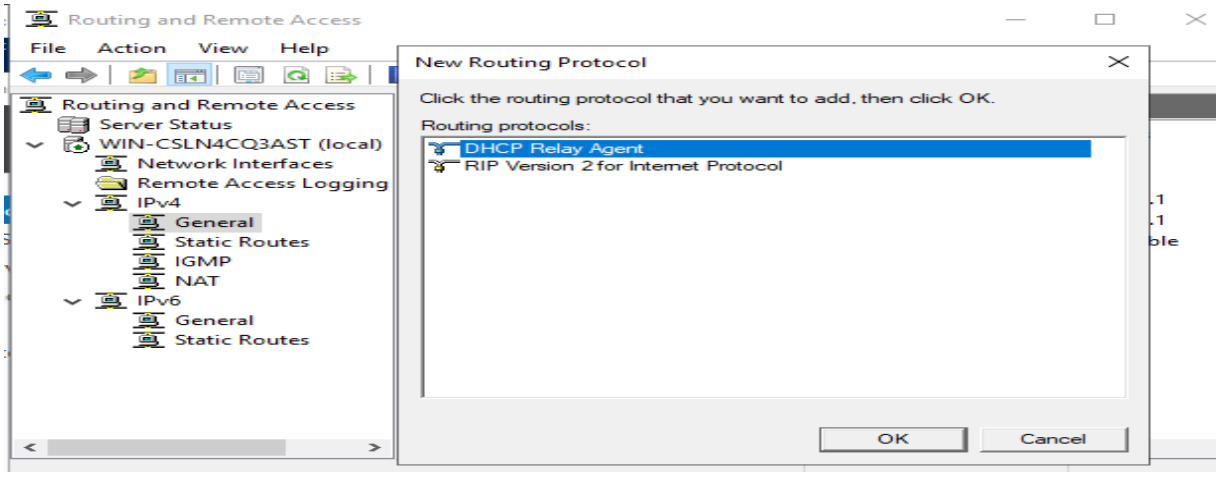
InterfaceAlias       : Ethernet
InterfaceIndex       : 5
InterfaceDescription : Intel(R) PRO/1000 MT Desktop Adapter
NetProfile.Name      : lakou.test
IPv4Address          : 192.168.10.10
IPv6DefaultGateway   :
IPv4DefaultGateway   : 192.168.10.1
DNSServer            : ::1
                       192.168.10.10

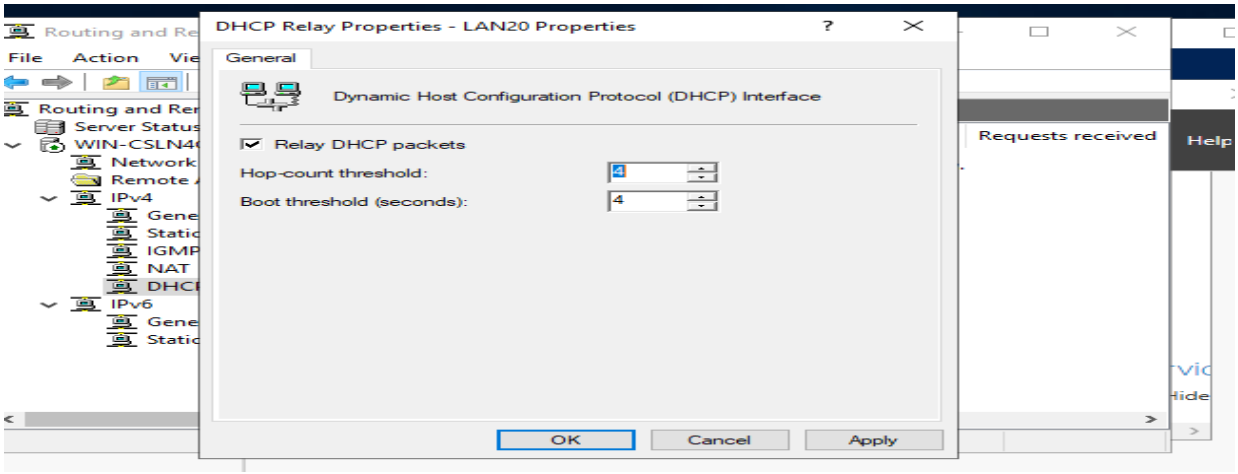

PS C:\Users\Administrator>
```

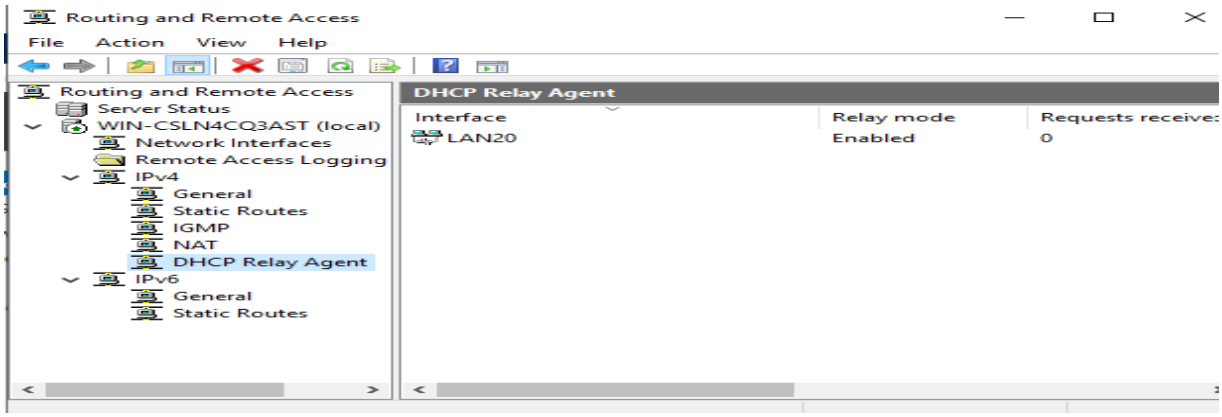
Activité 5 Configurer le relais DHCP

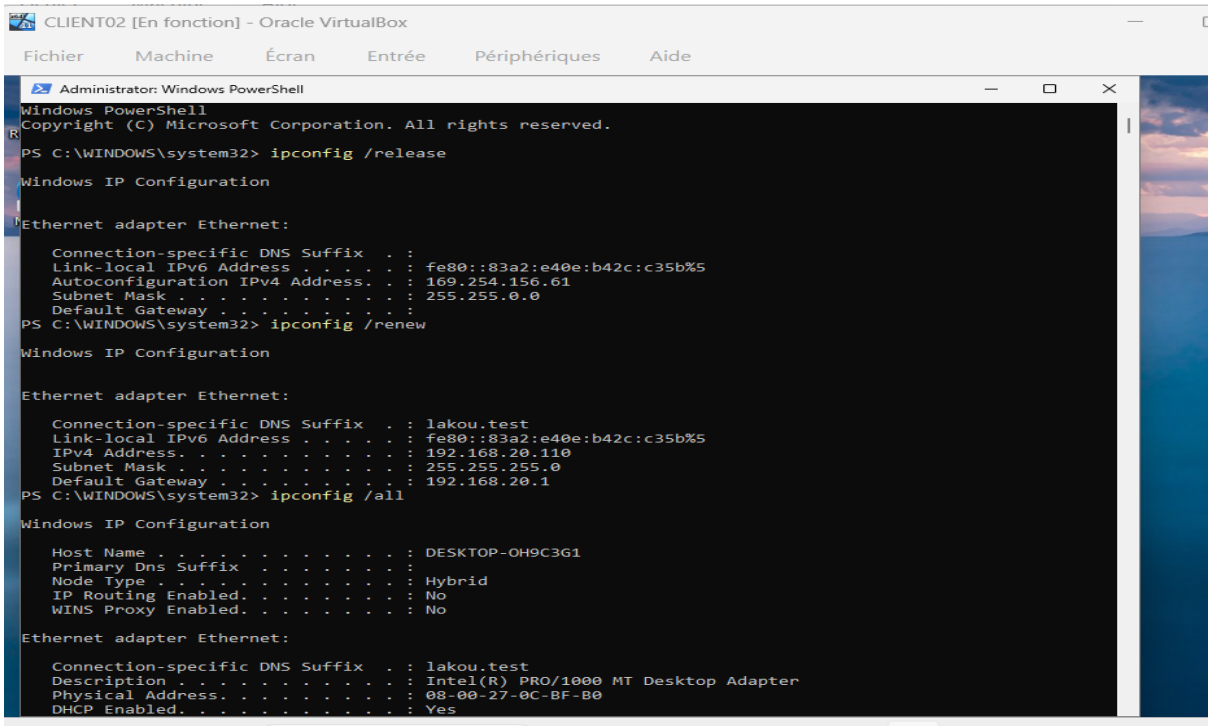


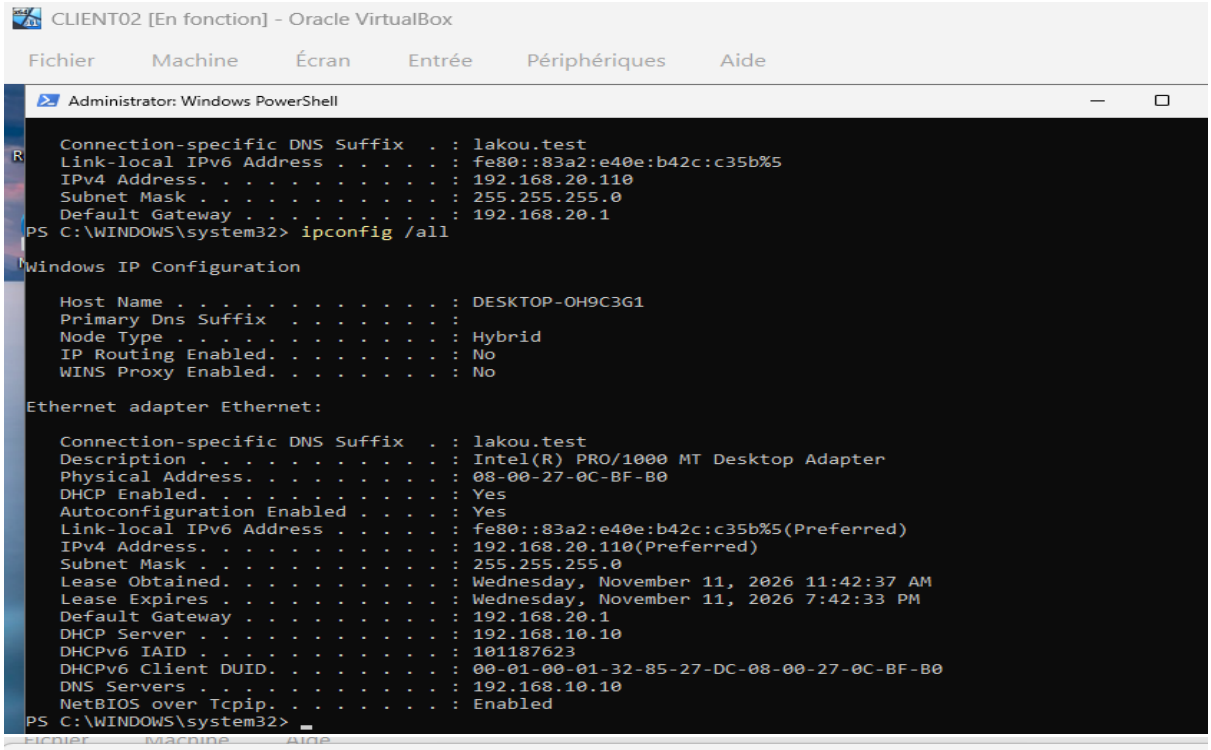


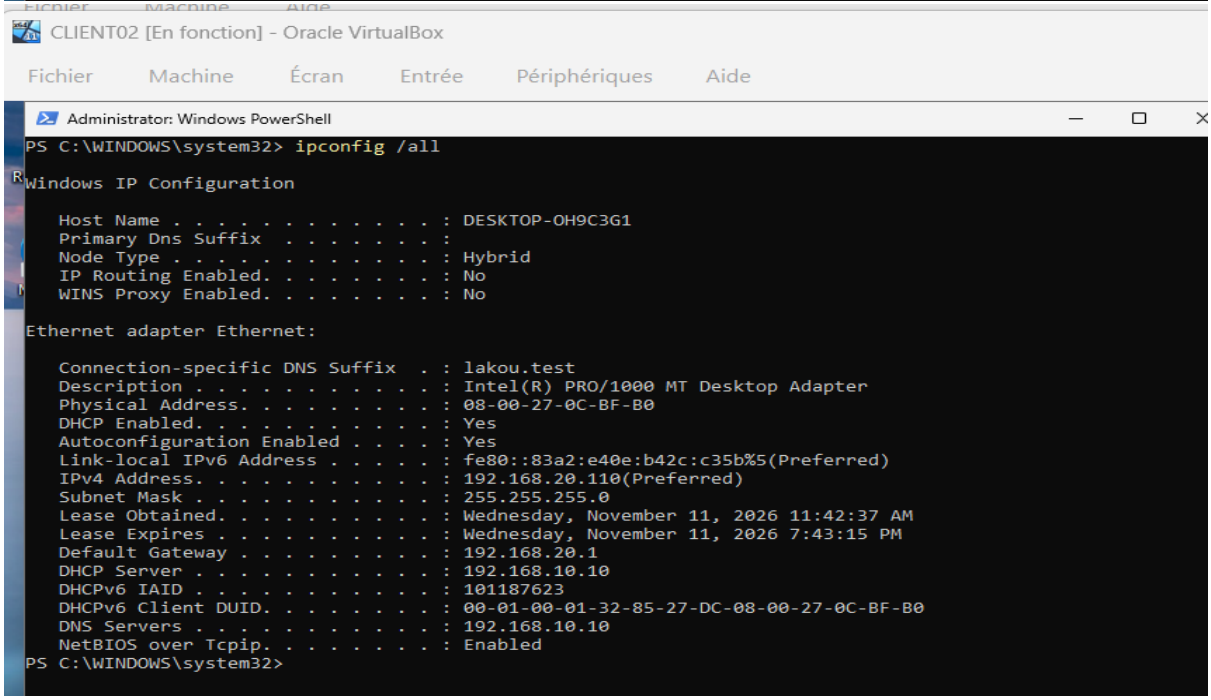

Routing and Remote Access
File Action View Help
Routing and Remote Access
Server Status
WIN-CSLN4CQ3AST (local)
Network Interfaces
Remote Access Logging
IPv4
General
Static Routes
IGMP
NAT
IPv6
General
Static Routes
New Routing Protocol
Click the routing protocol that you want to add, then click OK.
Routing protocols:
DHCP Relay Agent
RIP Version 2 for Internet Protocol
OK
Cancel

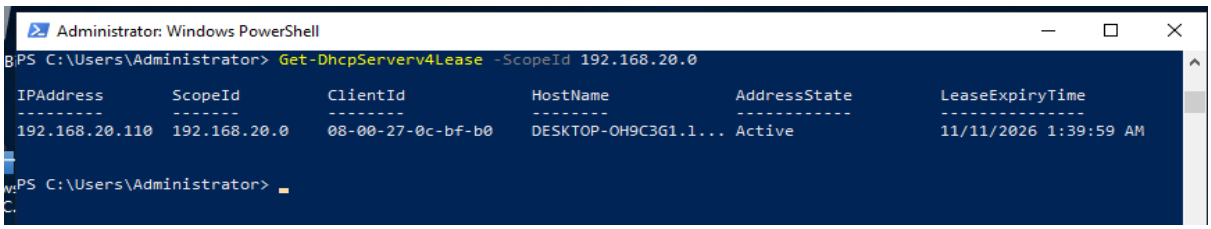

DHCP Relay Properties - LAN20 Properties
General
Dynamic Host Configuration Protocol (DHCP) Interface
Relay DHCP packets
Hop-count threshold:
4
Boot threshold (seconds):
4
OK
Cancel
Apply
Requests received
Help

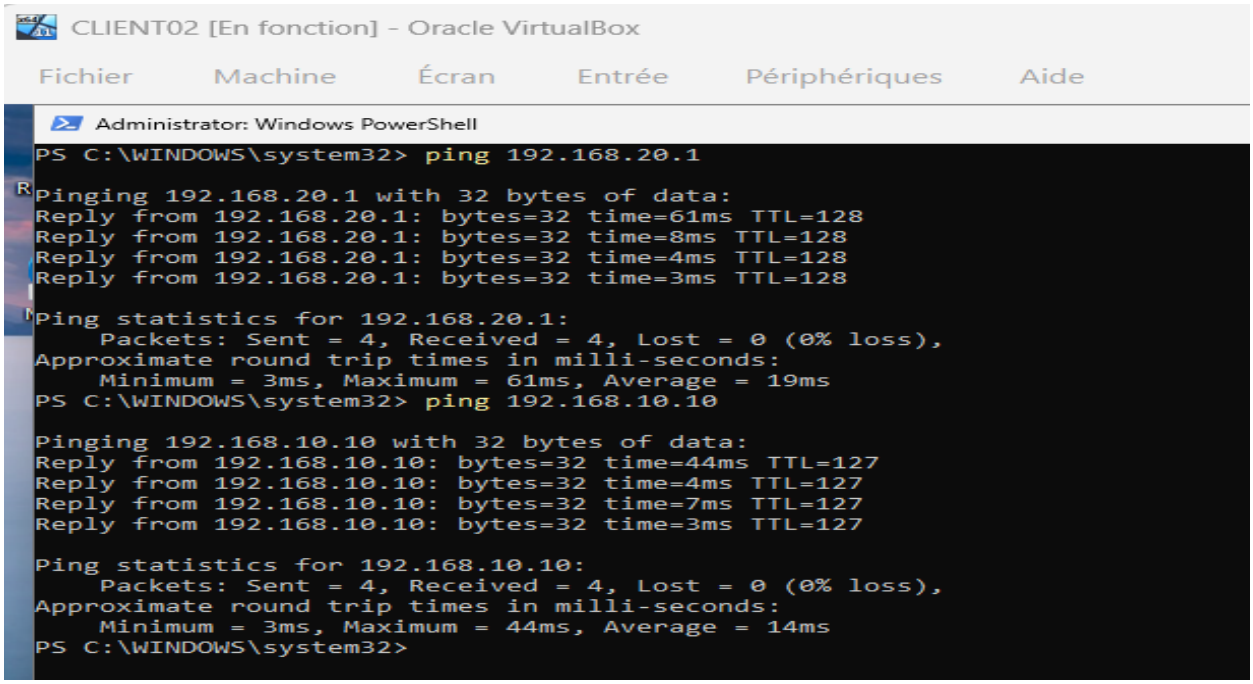

Routing and Remote Access
File Action View Help
Routing and Remote Access
Server Status
WIN-CSLN4CQ3AST (local)
Network Interfaces
Remote Access Logging
IPv4
General
Static Routes
IGMP
NAT
DHCP Relay Agent
IPv6
General
Static Routes
DHCP Relay Agent
Interface
Relay mode
Requests receive
LAN20
Enabled
0


CLIENT02 [En fonction] - Oracle VirtualBox
Fichier
Machine
Écran
Entrée
Périphériques
Aide
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\WINDOWS\system32> ipconfig /release
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::83a2:e40e:b42c:c35b%5
Autoconfiguration IPv4 Address. . : 169.254.156.61
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . :
PS C:\WINDOWS\system32> ipconfig /renew
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : lakou.test
Link-local IPv6 Address . . . . . : fe80::83a2:e40e:b42c:c35b%5
IPv4 Address. . . . . : 192.168.20.110
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.20.1
PS C:\WINDOWS\system32> ipconfig /all
Windows IP Configuration
Host Name . . . . . : DESKTOP-OH9C3G1
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : lakou.test
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-0C-BF-B0
DHCP Enabled. . . . . : Yes

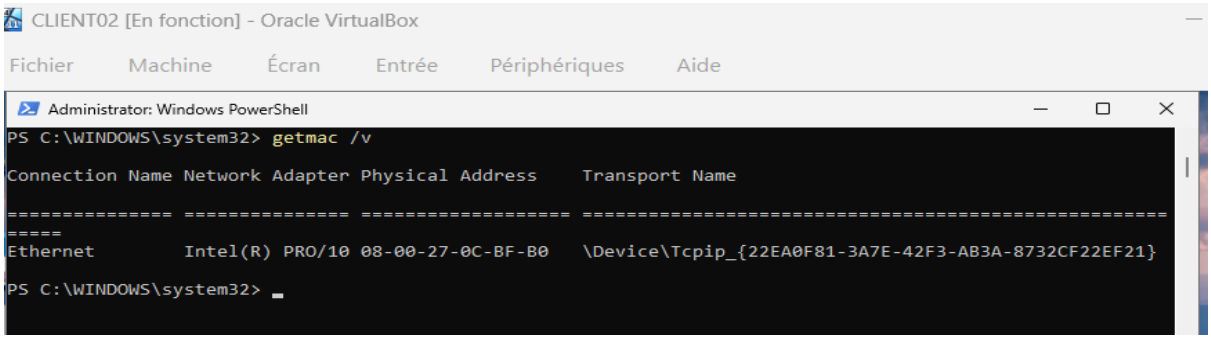

CLIENT02 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Administrator: Windows PowerShell
Connection-specific DNS Suffix . : lakou.test
Link-local IPv6 Address . . . . . : fe80::83a2:e40e:b42c:c35b%5
IPv4 Address. . . . . : 192.168.20.110
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.20.1
PS C:\WINDOWS\system32> ipconfig /all
Windows IP Configuration
Host Name . . . . . : DESKTOP-OH9C3G1
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : lakou.test
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-0C-BF-B0
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::83a2:e40e:b42c:c35b%5(Preferred)
IPv4 Address. . . . . : 192.168.20.110(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Wednesday, November 11, 2026 11:42:37 AM
Lease Expires . . . . . : Wednesday, November 11, 2026 7:42:33 PM
Default Gateway . . . . . : 192.168.20.1
DHCP Server . . . . . : 192.168.10.10
DHCPv6 IAID . . . . . : 101187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-32-85-27-DC-08-00-27-0C-BF-B0
DNS Servers . . . . . : 192.168.10.10
NetBIOS over Tcpip. . . . . : Enabled
PS C:\WINDOWS\system32>

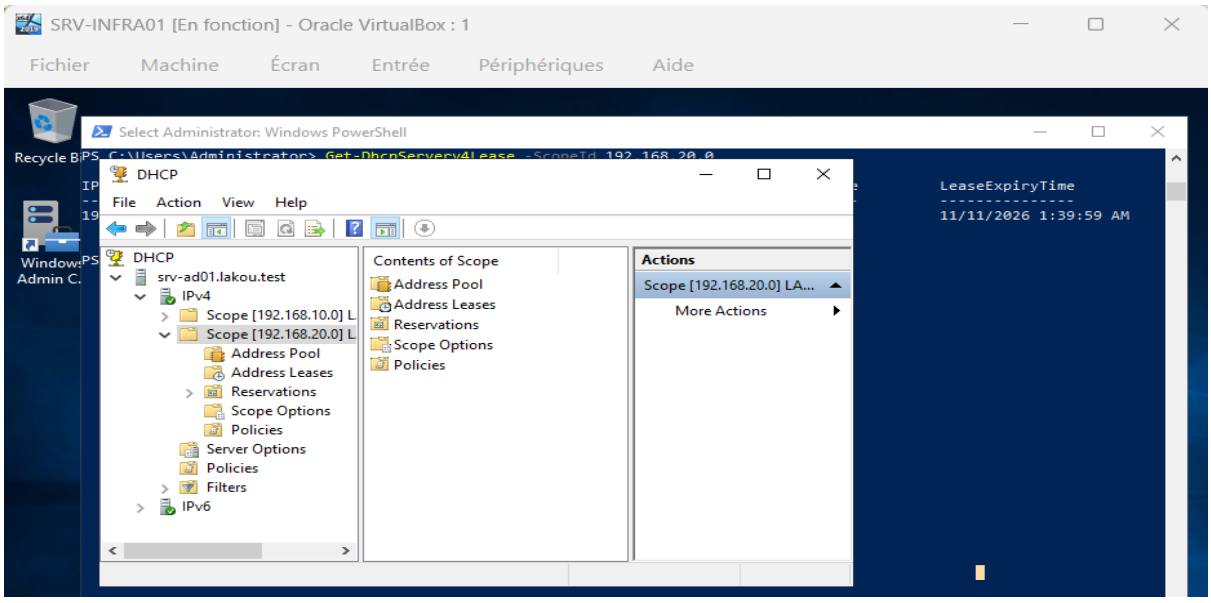

CLIENT02 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> ipconfig /all
Windows IP Configuration
Host Name . . . . . : DESKTOP-OH9C3G1
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : lakou.test
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-0C-BF-B0
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::83a2:e40e:b42c:c35b%5(Preferred)
IPv4 Address. . . . . : 192.168.20.110(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Wednesday, November 11, 2026 11:42:37 AM
Lease Expires . . . . . : Wednesday, November 11, 2026 7:43:15 PM
Default Gateway . . . . . : 192.168.20.1
DHCP Server . . . . . : 192.168.10.10
DHCPv6 IAID . . . . . : 101187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-32-85-27-DC-08-00-27-0C-BF-B0
DNS Servers . . . . . : 192.168.10.10
NetBIOS over Tcpip. . . . . : Enabled
PS C:\WINDOWS\system32>

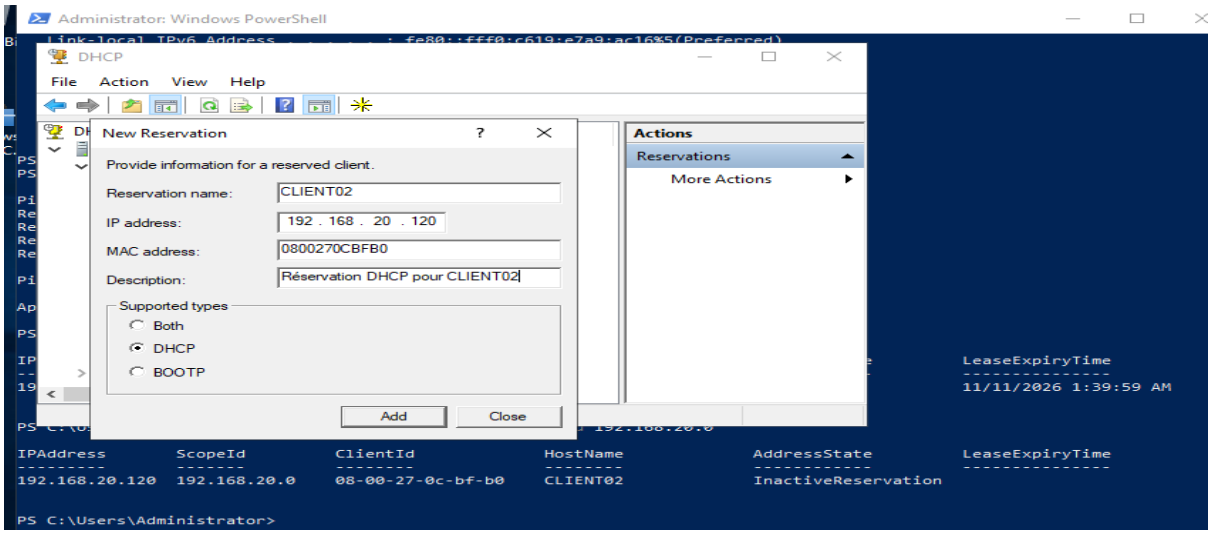

Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-DhcpServerv4Lease -ScopeId 192.168.20.0
IPAddress ScopeId ClientId HostName AddressState LeaseExpiryTime
192.168.20.110 192.168.20.0 08-00-27-0c-bf-b0 DESKTOP-OH9C3G1.1... Active 11/11/2026 1:39:59 AM
PS C:\Users\Administrator>

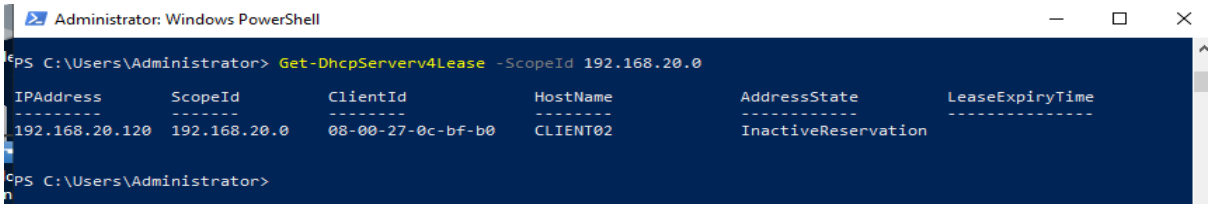


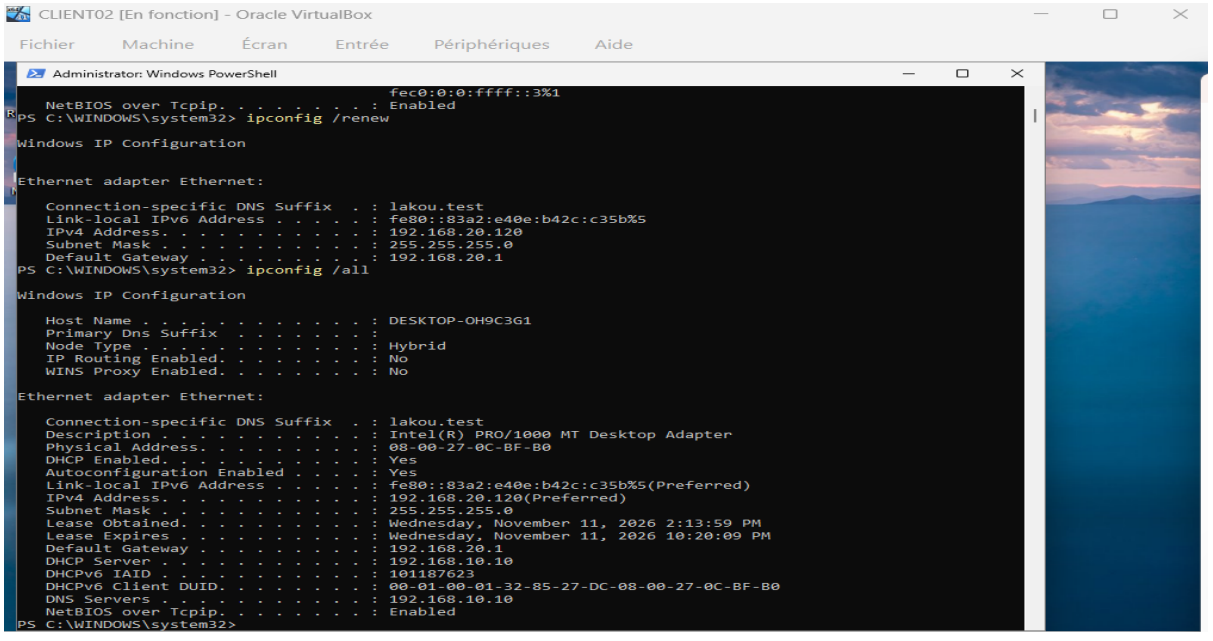
Activité 6 Créer une réservation DHCP

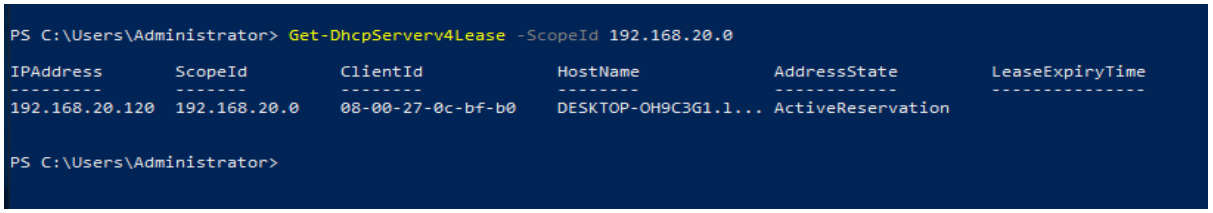




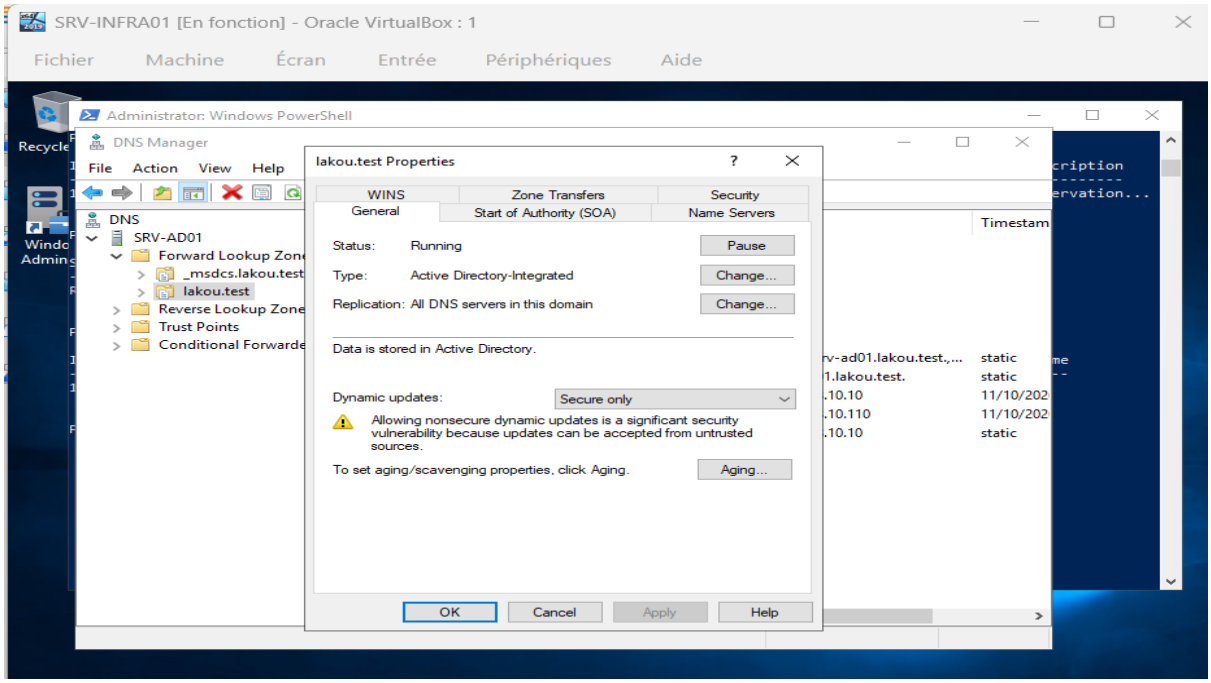

Administrator: Windows PowerShell
DHCP
File Action View Help
New Reservation
Provide information for a reserved client.
Reservation name: CLIENT02
IP address: 192 . 168 . 20 . 120
MAC address: 0800270CBFB0
Description: Réservation DHCP pour CLIENT02
Supported types
Both
DHCP
BOOTP
Add
Close
Actions
Reservations
More Actions
LeaseExpiryTime
11/11/2026 1:39:59 AM
IPAddress ScopeId ClientId HostName AddressState LeaseExpiryTime
192.168.20.120 192.168.20.0 08-00-27-0c-bf-b0 CLIENT02 InactiveReservation
PS C:\Users\Administrator>

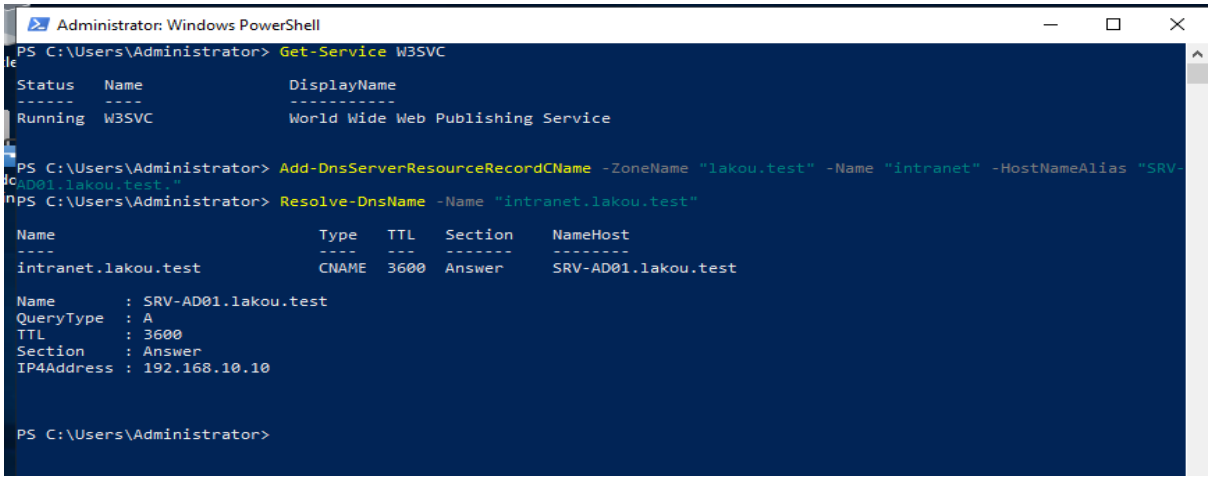

Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-DhcpServerv4Lease -ScopeId 192.168.20.0
IPAddress ScopeId ClientId HostName AddressState LeaseExpiryTime
192.168.20.120 192.168.20.0 08-00-27-0c-bf-b0 CLIENT02 InactiveReservation
PS C:\Users\Administrator>

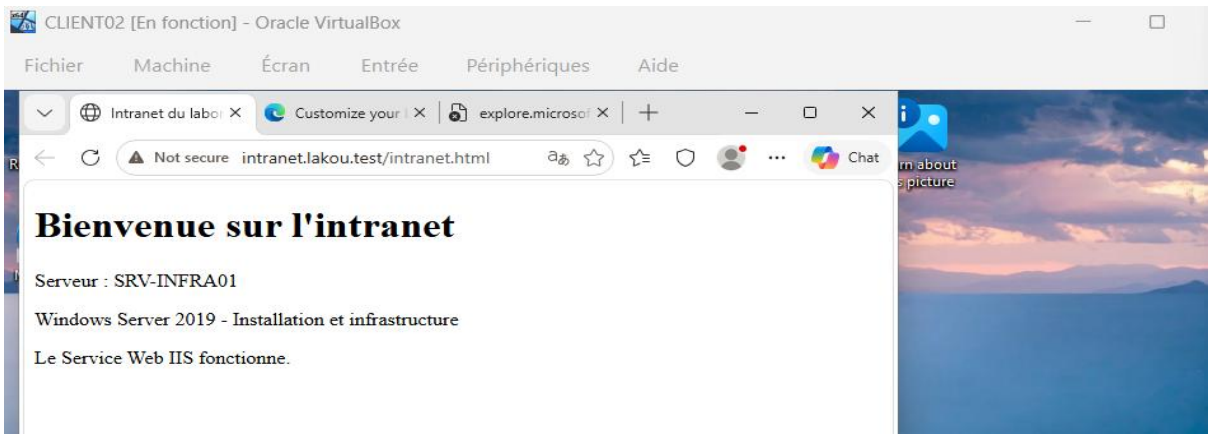

CLIENT02 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Administrator: Windows PowerShell
fec0:0:0:ffff::3%1
NetBIOS over Tcpip. . . . . : Enabled
PS C:\WINDOWS\system32> ipconfig /renew
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : lakou.test
Link-local IPv6 Address . . . . . : fe80::83a2:e40e:b42c:c35b%5
IPv4 Address. . . . . : 192.168.20.120
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.20.1
PS C:\WINDOWS\system32> ipconfig /all
Windows IP Configuration
Host Name . . . . . : DESKTOP-OH9C3G1
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : lakou.test
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-0C-BF-B0
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::83a2:e40e:b42c:c35b%5(Preferred)
IPv4 Address. . . . . : 192.168.20.120(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Wednesday, November 11, 2026 2:13:59 PM
Lease Expires . . . . . : Wednesday, November 11, 2026 10:20:09 PM
Default Gateway . . . . . : 192.168.20.1
DHCP Server . . . . . : 192.168.10.10
DHCPv6 IAID . . . . . : 101187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-32-85-27-DC-08-00-27-0C-BF-B0
DNS Servers . . . . . : 192.168.10.10
NetBIOS over Tcpip. . . . . : Enabled
PS C:\WINDOWS\system32>

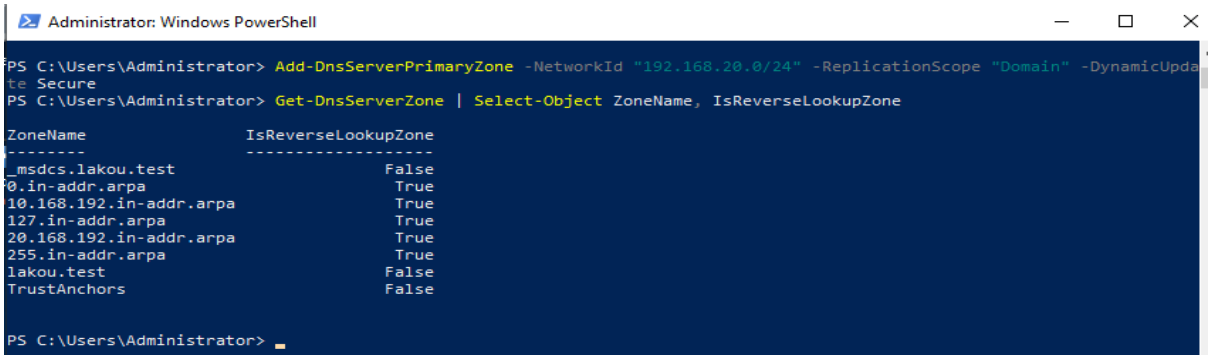

PS C:\Users\Administrator> Get-DhcpServerv4Lease -ScopeId 192.168.20.0
IPAddress ScopeId ClientId HostName AddressState LeaseExpiryTime
192.168.20.120 192.168.20.0 08-00-27-0c-bf-b0 DESKTOP-OH9C3G1.l... ActiveReservation
PS C:\Users\Administrator>

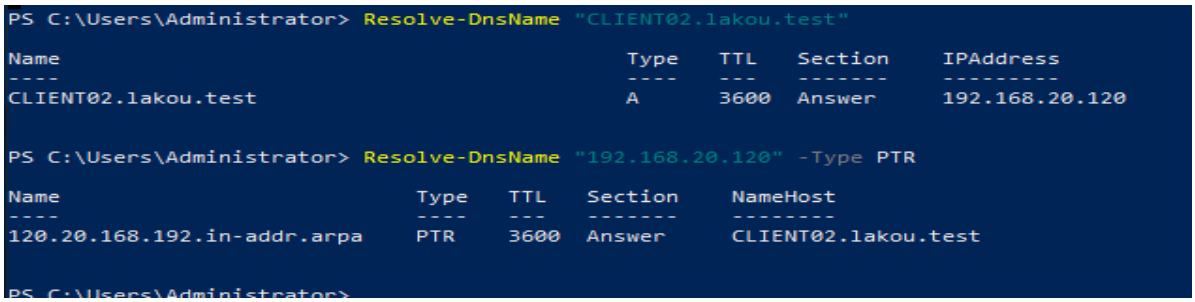
Activité 7 Configurer le DNS avancé

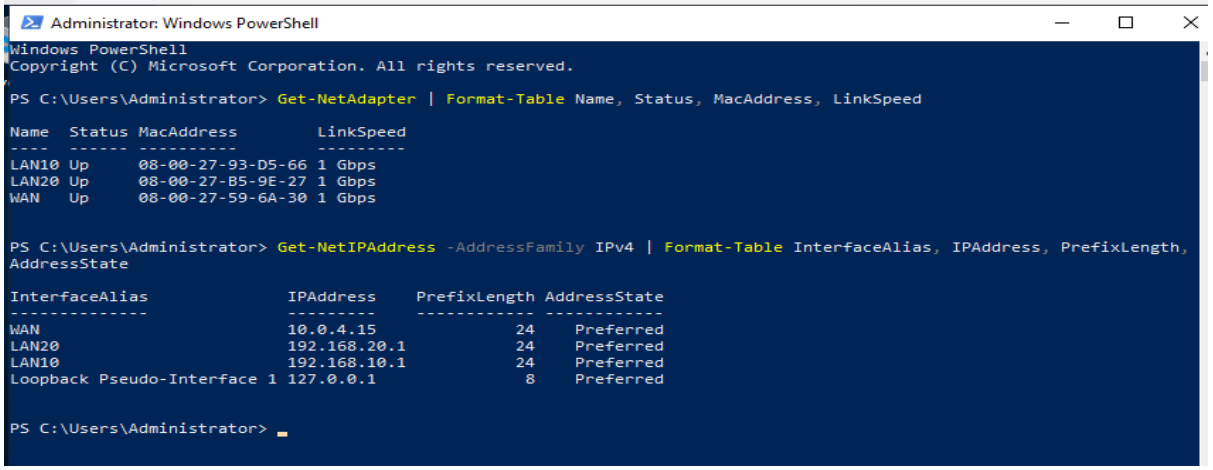


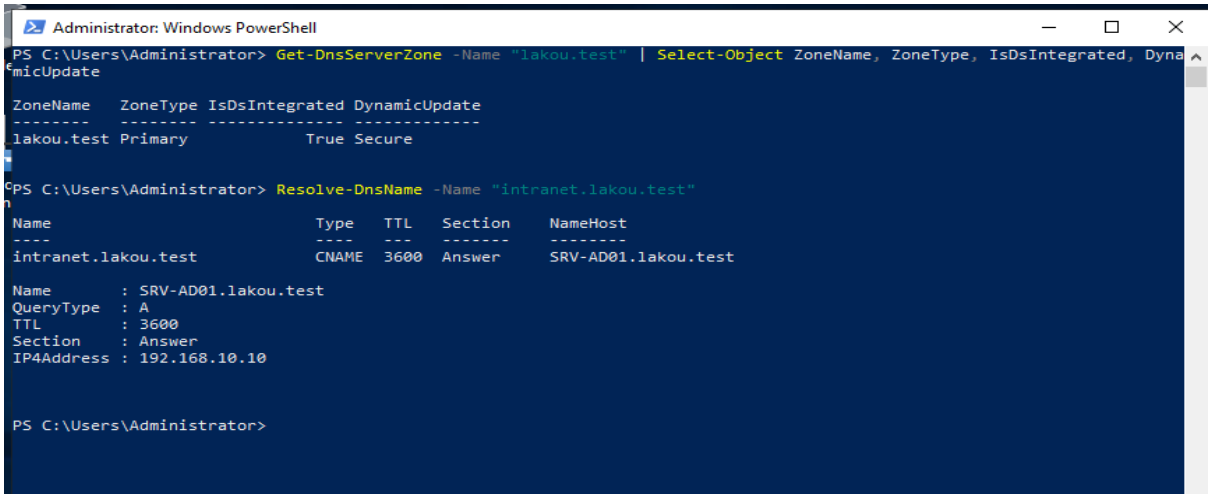


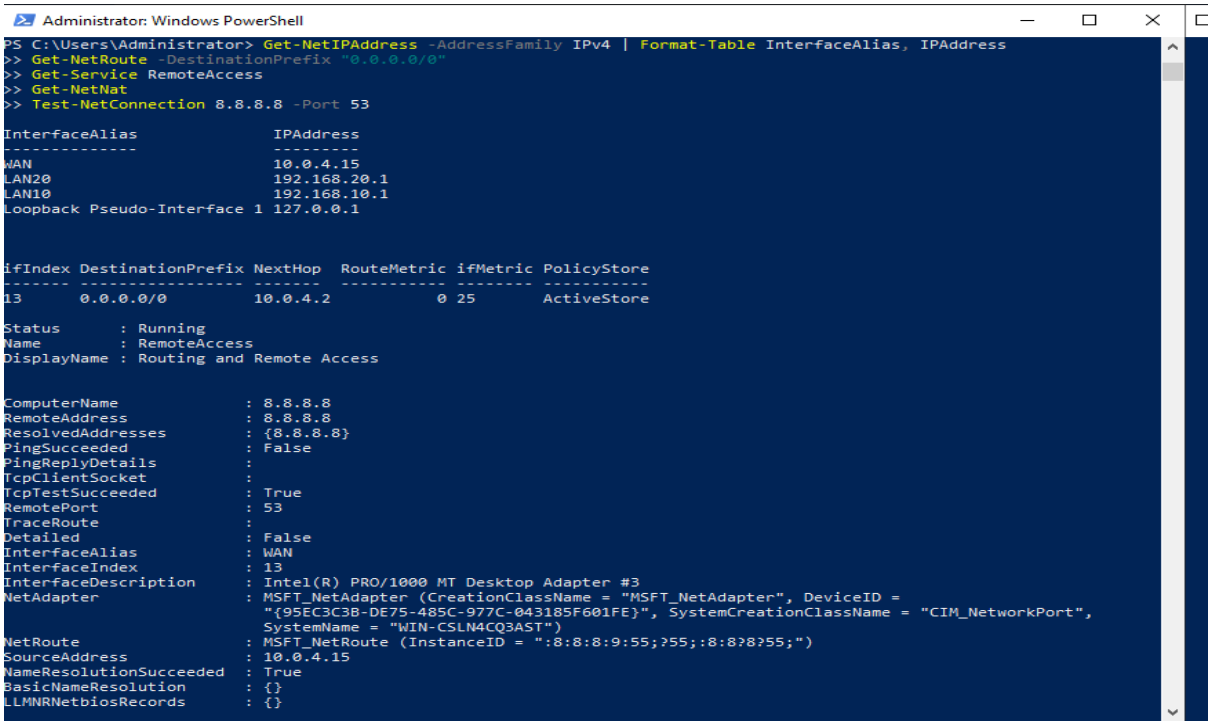


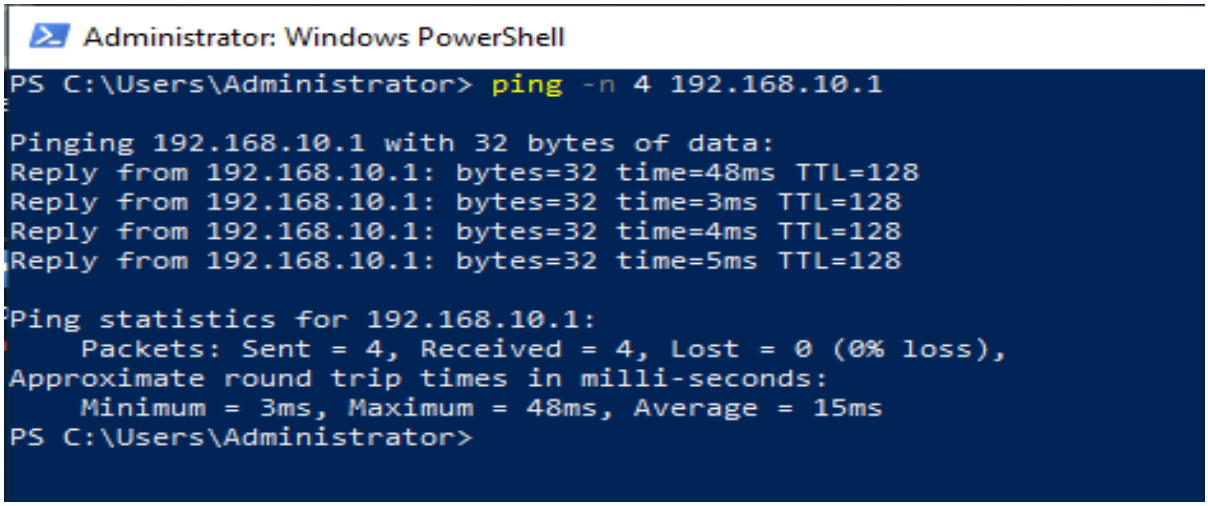

Administrator: Windows PowerShell
PS C:\Users\Administrator> Add-DnsServerPrimaryZone -NetworkId "192.168.20.0/24" -ReplicationScope "Domain" -DynamicUpdate Secure
PS C:\Users\Administrator> Get-DnsServerZone | Select-Object ZoneName, IsReverseLookupZone
ZoneName IsReverseLookupZone
_msdcs.lakou.test False
0.in-addr.arpa True
10.168.192.in-addr.arpa True
127.in-addr.arpa True
20.168.192.in-addr.arpa True
255.in-addr.arpa True
lakou.test False
TrustAnchors False
PS C:\Users\Administrator>

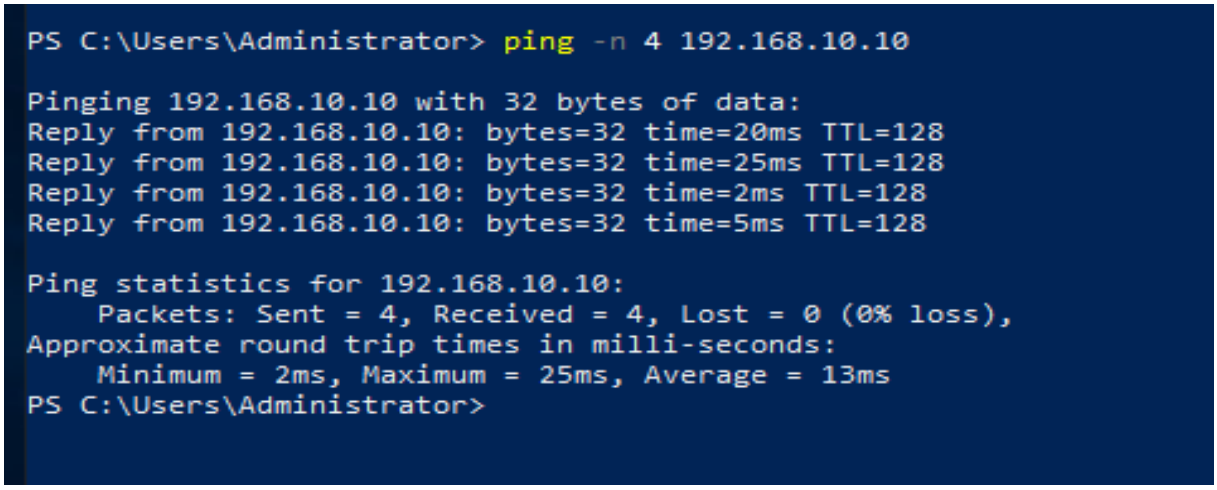

PS C:\Users\Administrator> Resolve-DnsName "CLIENT02.lakou.test"
Name Type TTL Section IPAddress
CLIENT02.lakou.test A 3600 Answer 192.168.20.120
PS C:\Users\Administrator> Resolve-DnsName "192.168.20.120" -Type PTR
Name Type TTL Section NameHost
120.20.168.192.in-addr.arpa PTR 3600 Answer CLIENT02.lakou.test


Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\Administrator> Get-NetAdapter | Format-Table Name, Status, MacAddress, LinkSpeed
Name Status MacAddress LinkSpeed
LAN10 Up 08-00-27-93-D5-66 1 Gbps
LAN20 Up 08-00-27-B5-9E-27 1 Gbps
WAN Up 08-00-27-59-6A-30 1 Gbps
PS C:\Users\Administrator> Get-NetIPAddress -AddressFamily IPv4 | Format-Table InterfaceAlias, IPAddress, PrefixLength, AddressState
InterfaceAlias IPAddress PrefixLength AddressState
WAN 10.0.4.15 24 Preferred
LAN20 192.168.20.1 24 Preferred
LAN10 192.168.10.1 24 Preferred
Loopback Pseudo-Interface 1 127.0.0.1 8 Preferred
PS C:\Users\Administrator>

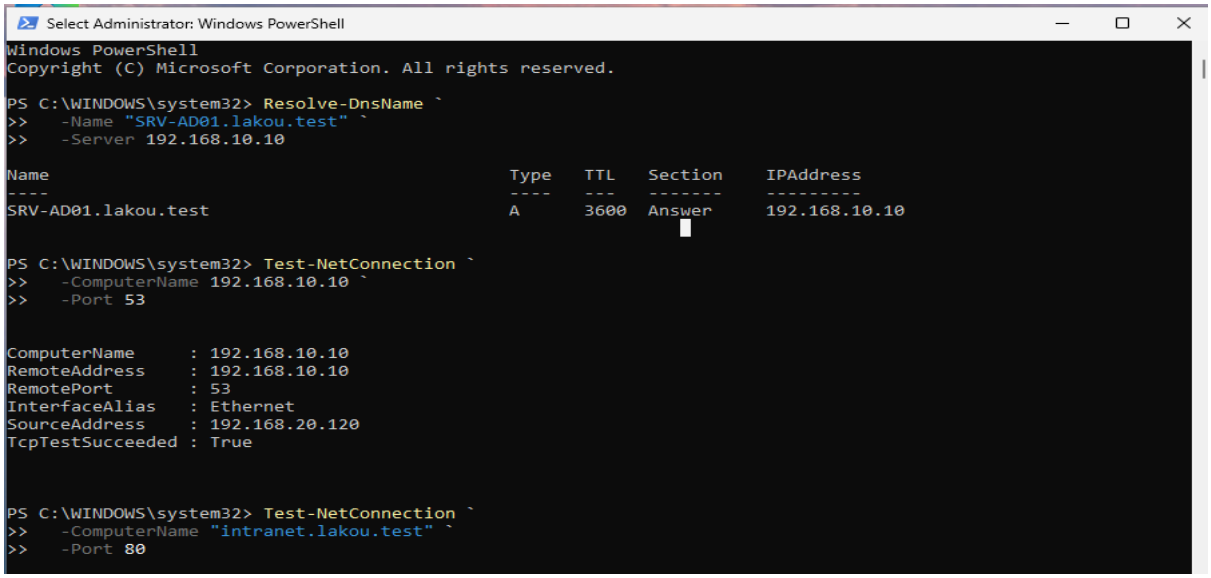

Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-DnsServerZone -Name "lakou.test" | Select-Object ZoneName, ZoneType, IsDsIntegrated, DynamicUpdate
ZoneName ZoneType IsDsIntegrated DynamicUpdate
lakou.test Primary True Secure
PS C:\Users\Administrator> Resolve-DnsName -Name "intranet.lakou.test"
Name Type TTL Section NameHost
intranet.lakou.test CNAME 3600 Answer SRV-AD01.lakou.test
Name : SRV-AD01.lakou.test
QueryType : A
TTL : 3600
Section : Answer
IP4Address : 192.168.10.10
PS C:\Users\Administrator>

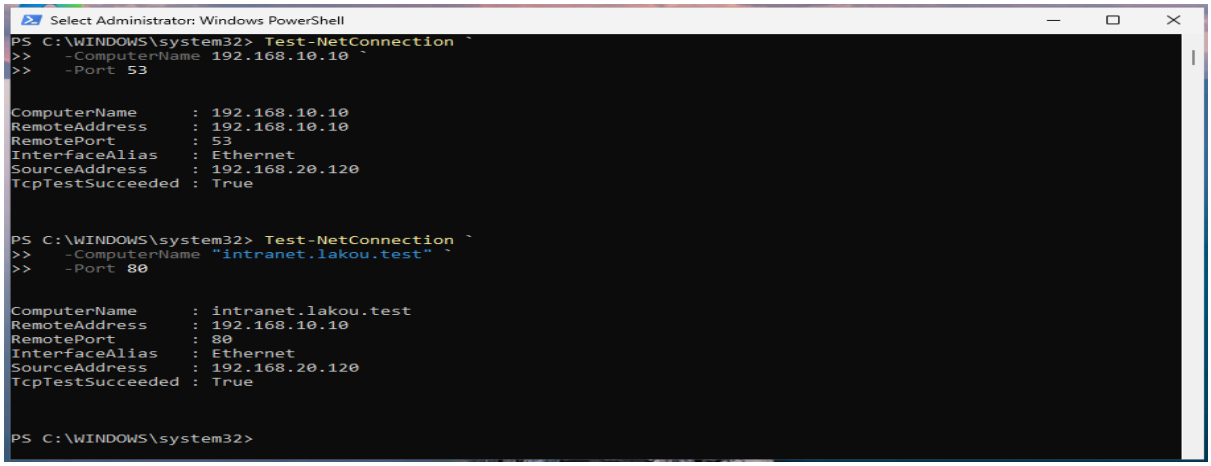

Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-NetIPAddress -AddressFamily IPv4 | Format-Table InterfaceAlias, IPAddress
>> Get-NetRoute -DestinationPrefix "0.0.0.0/0"
>> Get-Service RemoteAccess
>> Get-NetNat
>> Test-NetConnection 8.8.8.8 -Port 53
InterfaceAlias IPAddress
WAN 10.0.4.15
LAN20 192.168.20.1
LAN10 192.168.10.1
Loopback Pseudo-Interface 1 127.0.0.1
ifIndex DestinationPrefix NextHop RouteMetric ifMetric PolicyStore
13 0.0.0.0/0 10.0.4.2 0 25 ActiveStore
Status : Running
Name : RemoteAccess
DisplayName : Routing and Remote Access
ComputerName : 8.8.8.8
RemoteAddress : 8.8.8.8
ResolvedAddresses : {8.8.8.8}
PingSucceeded : False
PingReplyDetails :
TcpClientSocket :
TcpTestSucceeded : True
RemotePort : 53
TraceRoute :
Detailed : False
InterfaceAlias : WAN
InterfaceIndex : 13
InterfaceDescription : Intel(R) PRO/1000 MT Desktop Adapter #3
NetAdapter : MSFT_NetAdapter (CreationClassName = "MSFT_NetAdapter", DeviceID = "{95EC3C3B-DE75-485C-977C-043185F601FE}", SystemCreationClassName = "CIM_NetworkPort", SystemName = "WIN-CSLN4CQ3AST")
NetRoute : MSFT_NetRoute (InstanceID = ":8:8:8:9:55;?55;:8:8?8?55;")
SourceAddress : 10.0.4.15
NameResolutionSucceeded : True
BasicNameResolution : {}
LLMNRNetbiosRecords : {}

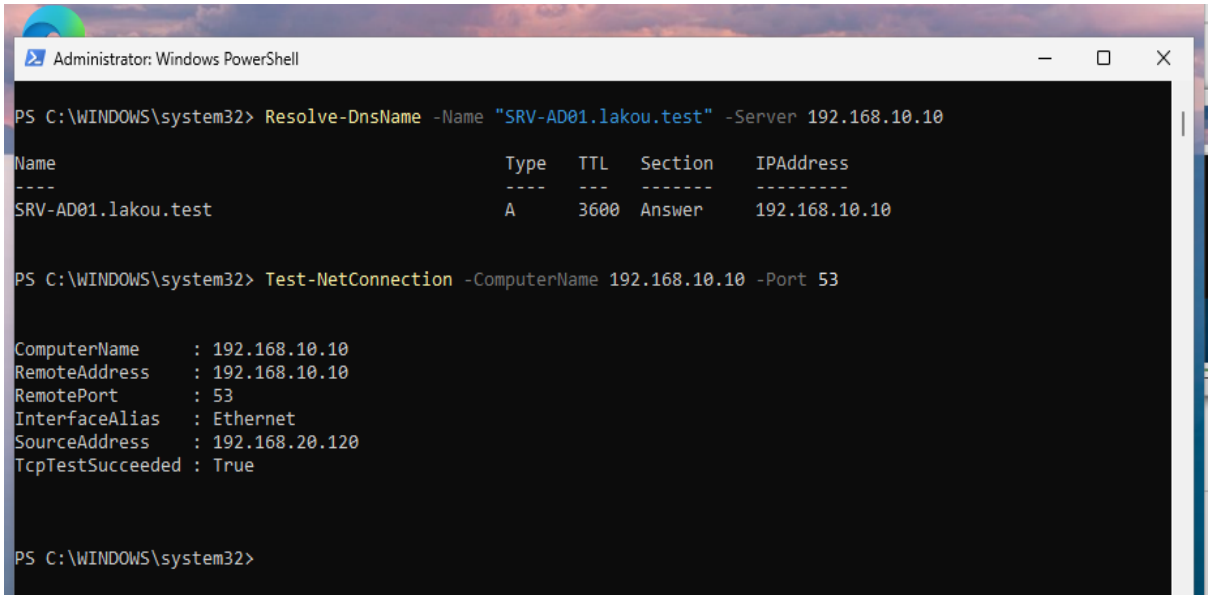

Administrator: Windows PowerShell
PS C:\Users\Administrator> ping -n 4 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=48ms TTL=128
Reply from 192.168.10.1: bytes=32 time=3ms TTL=128
Reply from 192.168.10.1: bytes=32 time=4ms TTL=128
Reply from 192.168.10.1: bytes=32 time=5ms TTL=128
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 48ms, Average = 15ms
PS C:\Users\Administrator>

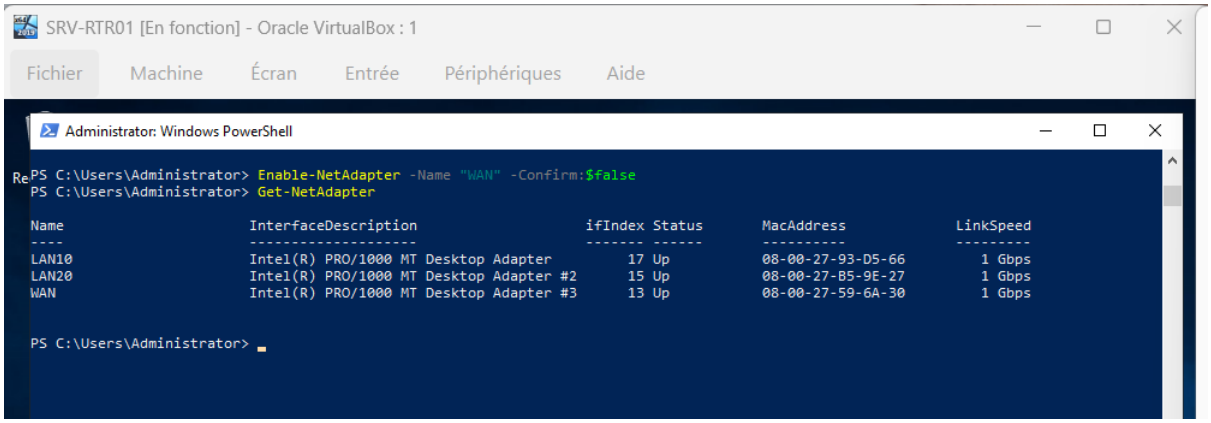

PS C:\Users\Administrator> ping -n 4 192.168.10.10
Pinging 192.168.10.10 with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time=20ms TTL=128
Reply from 192.168.10.10: bytes=32 time=25ms TTL=128
Reply from 192.168.10.10: bytes=32 time=2ms TTL=128
Reply from 192.168.10.10: bytes=32 time=5ms TTL=128
Ping statistics for 192.168.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 25ms, Average = 13ms
PS C:\Users\Administrator>

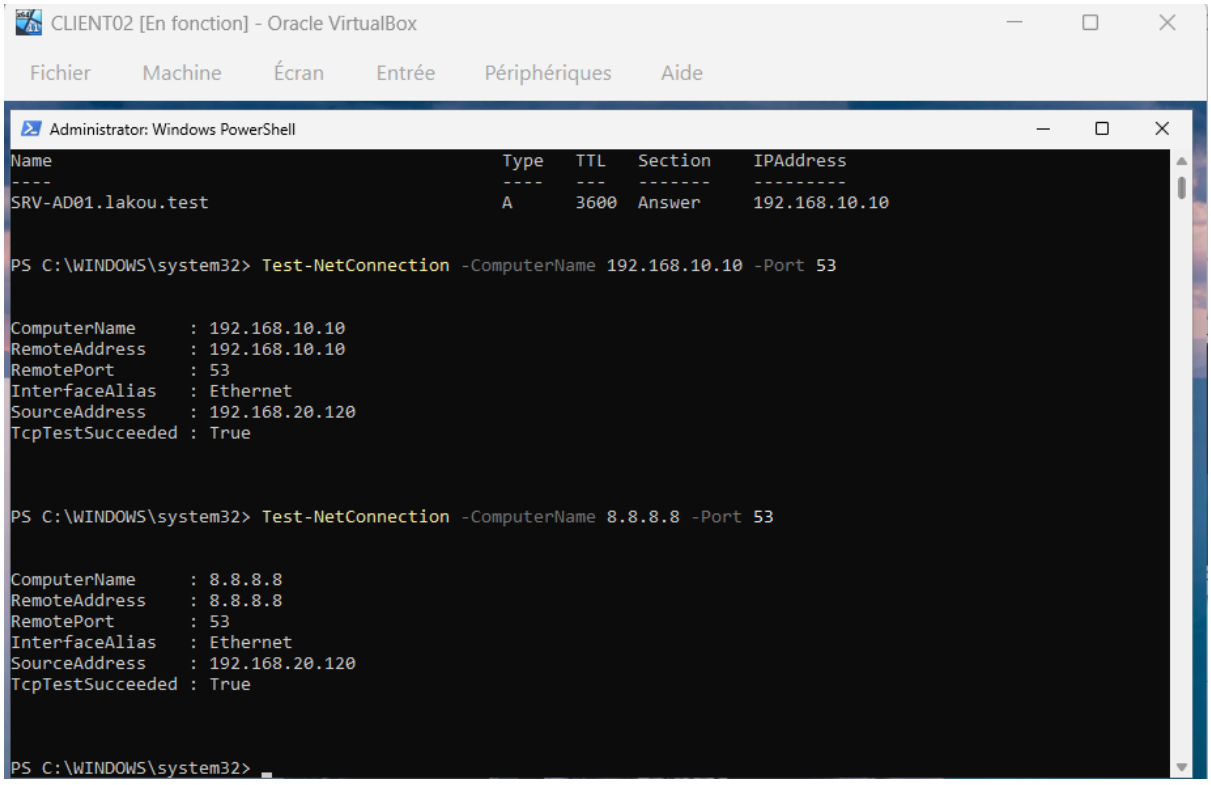
Activité 8 Validation du routage et du NAT











Conclusion

Après la réactivation de l'interface WAN, les trois interfaces réseau (LAN10, LAN20 et WAN) affichent l'état Up. Les services internes ont également été accessibles pendant l'interruption de la connexion extérieure. Cette manipulation confirme le maintien du fonctionnement du réseau local malgré une coupure du WAN.

Questions

1. Quelles sont les quatre étapes classiques de DHCP ?

Les quatre étapes classiques de DHCP (DORA)

- **Discover** : le client diffuse en broadcast pour chercher un serveur DHCP.
- **Offer** : le serveur propose une adresse IP avec ses paramètres (masque, passerelle, DNS).
- **Request** : le client demande officiellement l'adresse proposée.
- **Acknowledge** : le serveur confirme et accorde le bail.

2. Pourquoi utiliser un relais DHCP ?

Le broadcast DHCP ne traverse pas les routeurs. Le relais (DHCP Relay / `ip helper-address`) capte la requête du client et la transmet en unicast au serveur DHCP situé dans un autre réseau. On évite ainsi d'installer un serveur DHCP dans chaque sous-réseau.

3. Comment le serveur choisit-il l'étendue du client distant ?

Le relais ajoute dans la requête son adresse **giaddr** (l'adresse de son interface côté client). Le serveur regarde ce giaddr, trouve l'étendue (scope) qui correspond à ce sous-réseau, puis attribue une adresse de cette étendue.

4. Quelle différence existe-t-il entre routage et NAT ?

- **Routage** : acheminer les paquets d'un réseau à un autre selon la table de routage, sans modifier les adresses IP.
- **NAT** : traduire les adresses IP (généralement privées vers publique) pendant le passage du paquet.
  Le routage décide *où* envoyer le paquet ; le NAT *modifie* ses adresses.

5. Deux VLAN peuvent-ils communiquer automatiquement ?

Non. Chaque VLAN est un réseau séparé (domaine de broadcast distinct). Pour qu'ils communiquent, il faut un équipement de niveau 3 : un routeur (router-on-a-stick) ou un switch L3 avec routage inter-VLAN.

6. La segmentation interdit-elle toute communication entre services ?

Non. Elle sépare les réseaux pour réduire les risques et mieux contrôler les flux, mais la communication reste possible si elle est autorisée par des règles (ACL, pare-feu). On n'autorise que le trafic nécessaire.

7. À quoi sert un alias CNAME ?

Un CNAME est un enregistrement DNS qui fait pointer un nom vers un autre nom (pas vers une IP). Exemple : `www.exemple.ht` → `exemple.ht`. Si l'IP du nom cible change, l'alias suit automatiquement.

8. Le NAT remplace-t-il un pare-feu ?

Non. Le NAT cache les adresses internes et bloque indirectement les connexions entrantes non sollicitées, mais ce n'est pas sa fonction. Il ne filtre pas le trafic selon des règles de sécurité, n'inspecte pas les paquets et ne protège pas contre les attaques. Un pare-feu reste nécessaire.